

Advanced Computer Networks

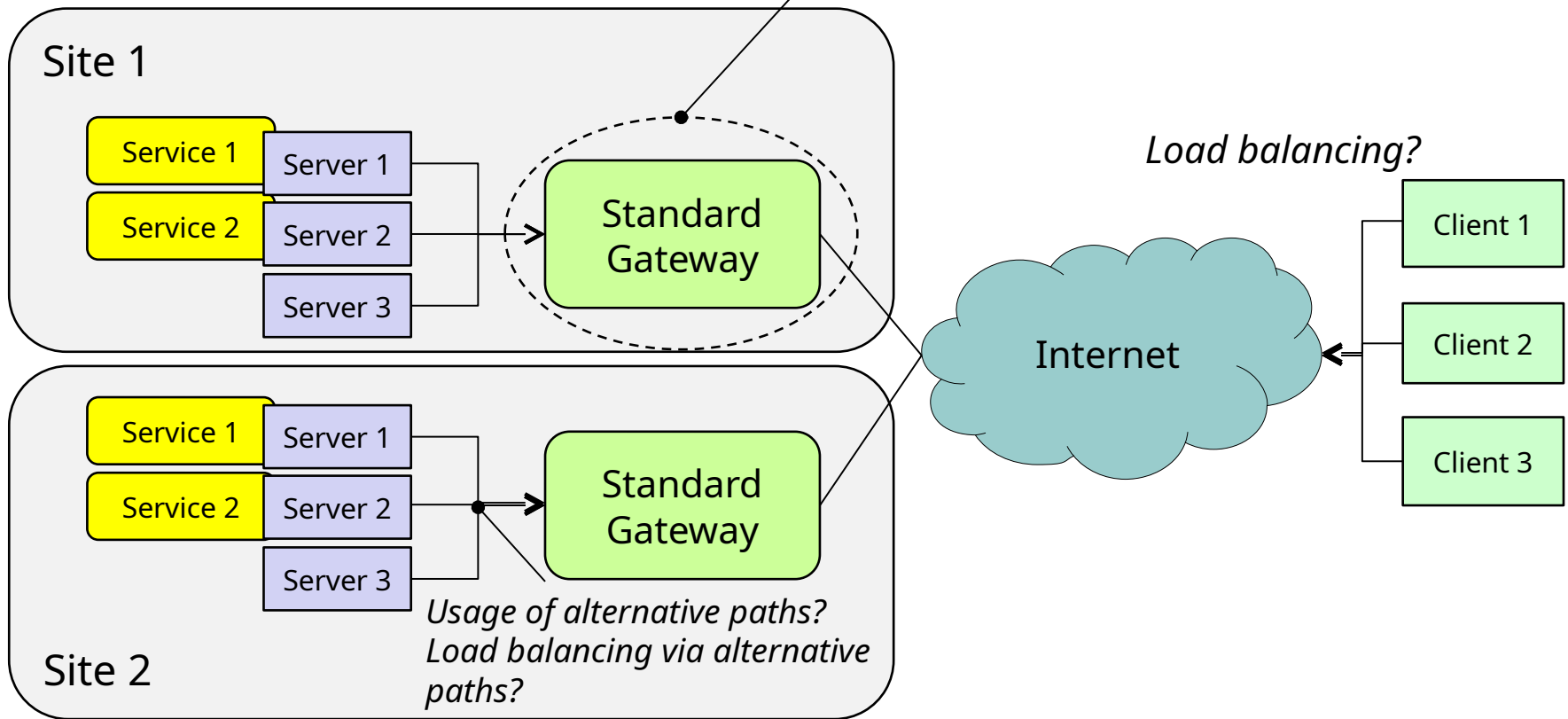
Redundancy, High Availability and Load Balancing

Winter term 2025/26

marius.feldmann@tu-dresden.de

Introduction

*Avoidance of a
Single Point of Failure (SPoF)*



- Lecture addresses basic approaches
 - Redundant deployment of services as well as load-balancing between services is not addressed (see e.g. "HAProxy")
 - "Classical" approaches for L2 redundancy such as the Spanning Tree Protocol (STP) or Transparent Interconnection of Lots of Links (TRILL) are not discussed (as they are not relevant in the case of using mechanisms on L3)

Outline

1. Selected terms
2. Round-robin DNS
3. WAN load balancing
 - Overview
 - Example based on VyOs
4. Link aggregation
 - Overview
 - Example based on Linux
5. Equal-Cost Multi-Path routing (ECMP)
 - Overview
 - ECMP using OSPF
 - ECMP in a Clos network topology
6. Virtual Router Redundancy Protocol (VRRP)
 - Overview
 - Selected protocol details
 - Discussion of VRRP for load balancing
7. Overview of stateful failover approach

Selected terms

- **Availability:**
 - Quantified based on the amount of time the network can be used to fulfil a desired goal
 - Typically, specified as a percentage value
 - 99.999%: ("five nines") = maximum of 5.256 mins. down-time per year
 - 99.99999%: ("seven nines") = maximum of 3,15 secs. down-time per year
 - Desired availability is often specified in **Service Level Agreements** (SLAs) (e.g. as part of the contract between ISP and customer)
 - SLAs usually mention penalty if the availability goals are not fulfilled
- **Mean Time Between Failures (MTBF):**
 - Average time the network works as expected (no error state given)
- **Mean Time To Repair (MTTR):**
 - Average time used to repair a network in case of an error event

$$\text{Availability} = \frac{\text{MTBF}}{\text{MTBF} + \text{MTTR}}$$

Selected terms

- Fault-tolerant and robust behavior is achieved via **redundancy** of critical network components
- **Load balancing:** Distribution of requests to redundant, parallelly active systems
 - Differentiation of:
 - *Flow-based* load balancing: For each flow / connection, an interface is selected typically used for the whole duration of the flow's existence
 - *Packet-based* load balancing: An interface is selected for each packet
- **Failover:** Activation of an alternative/redundant system in order to take control of the tasks of the failed system / the system that can not fulfil the specified QoS parameters (temporarily)
- Goal of
 - Failover approach: Select and provide alternative systems in the case of a failure or in the case QoS parameters can not be fulfilled by a primary system (active/passive setup)
 - Load balancing approach: Redundantly deployed resources are leveraged in parallel (active/active setup)

Selected approaches

Round Robin DNS

- Mapping of a domain name to several IP addresses to which a DNS request may be forwarded
-

WAN load balancing

- Load balancing between several WAN interfaces (preferably connected to different ISPs with separate infrastructure)
-

Link aggregation / Bonding

- Aggregation of several dedicated Ethernet interfaces to a logical one
-

Equal-Cost Multi-Path Routing

- Usage of several redundant entries in the forwarding information base (FIB) for load balancing purposes
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Virtual Router Redundancy Protocol

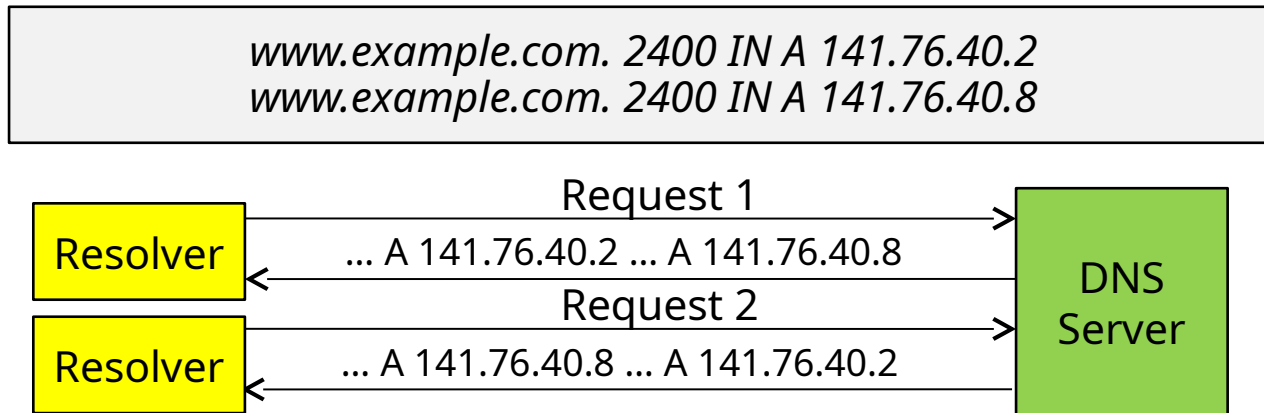
- Forwarding is done via a virtual IP address that can be assigned to different router instances
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Stateful Failover

- State managed during a connection is transparently migrated to a backup system in the case of a failover event

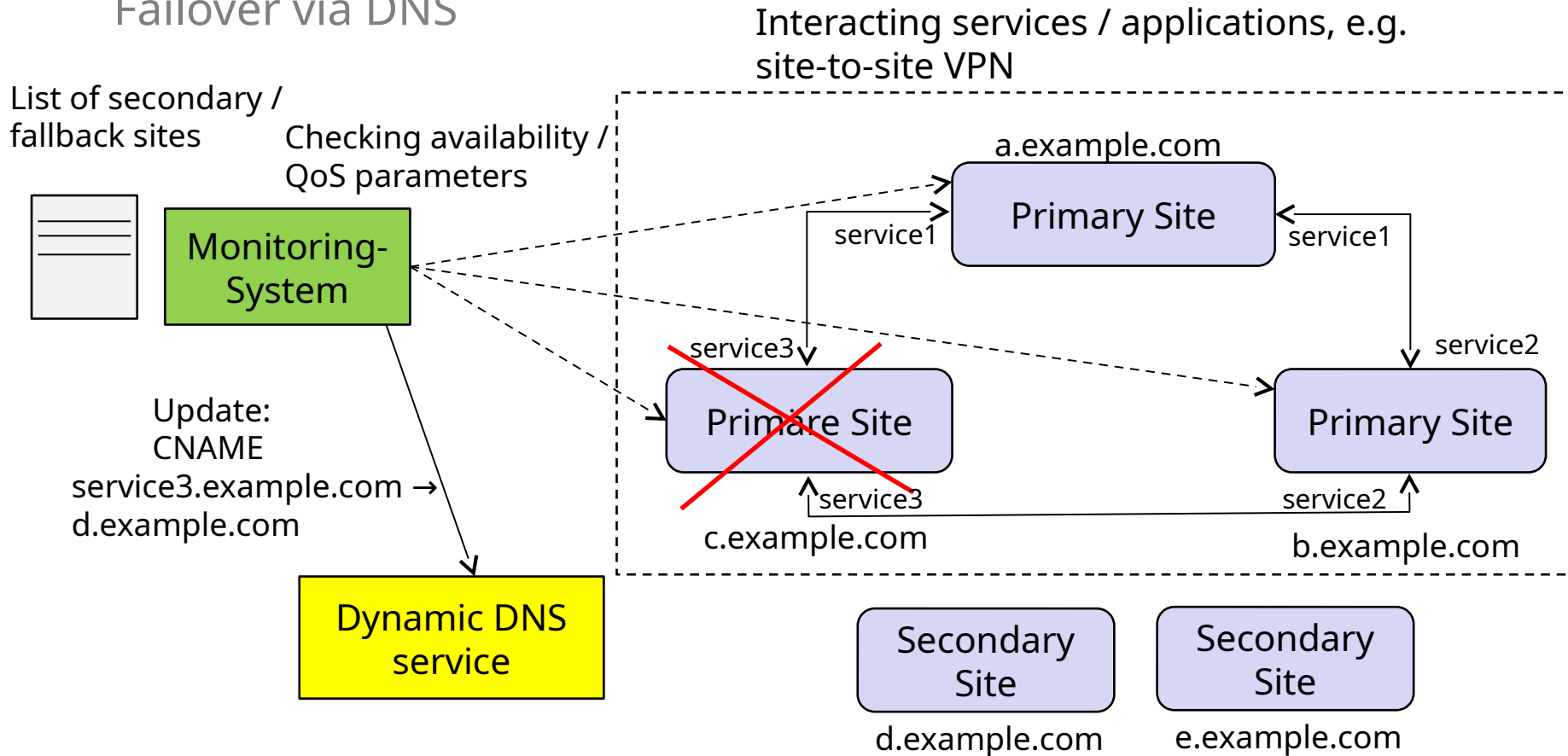
Round-robin DNS

- Domain Name System makes it possible to specify several Resource Records having the same class and type information and different resource data for a single domain name
- DNS server returns entries in different order → clients usually select the first entry resulting in load balancing e.g. between specified IP addresses



- Potential order of the returned resource records can be typically configured on the DNS server, e.g. in the case of the BIND DNS server:
 - **Fixed:** Order used in the zone file is used
 - **Random:** Random order
 - **Cyclic:** Round robin (default value)
- Disadvantages:
 - Load balancing does not take server load into account
 - Approach is bypassed by caching

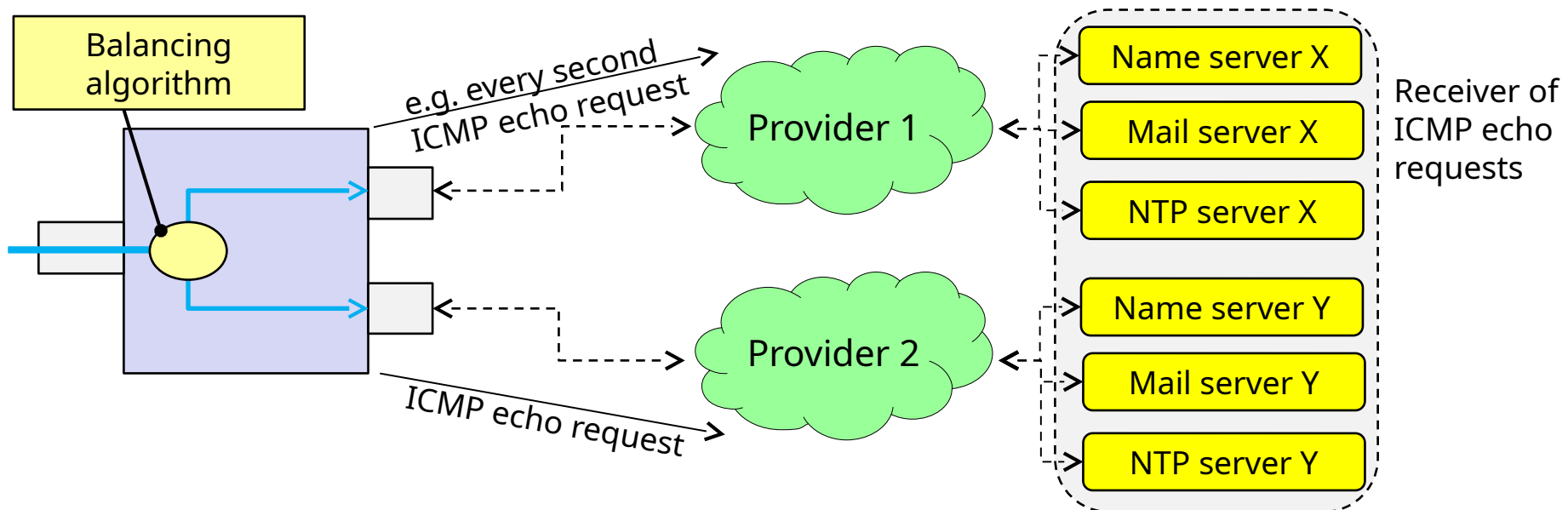
Failover via DNS



- **Sketched approach:**
 - Monitoring system checks collaborating services, which reference each other via names resolved via CNAME DNS entries
 - As soon as one of the systems/services crashes, the DNS entry is updated in order to reference to an alternative / backup system
 - Monitoring system has to be deployed in a redundant manner plus a quorum-based approach should be applied if a monitoring system detects that a system/service is not available

WAN load balancing

- Refers to a router / gateway setup with several uplink Internet / WAN interfaces ("multi-homing") leveraged for scheduling egress packets
- In an ideal setup, interfaces offer paths via ISPs which are independent from each other
- Load balancing may use health checks to validate if Internet / WAN connectivity works as expected → in the case of a failed check, an interface will be excluded from load balancing



- Efficient load balancing algorithm: (Weighted) Round-robin / (Weighted) random distribution load balancing

WAN load balancing – Example: VyOs

```
wan{
  interface-health eth0{
    failure-count 5
    nexthop 192.168.0.1
    test 5 {
      target 9.9.9.9
      type ping
    }
    test 10 {...}
  }
  interface-health eth1{
    failure-count 5
    nexthop dhcp
    test 5 { target mail.tu-dresden.de
      type ping
    }
    test 10 {...}
  }
  rule 5 {
    failover
    inbound-interface eth2
    interface eth0 { weight 1
    }
    interface eth1 { weight 10
    }
  }
}
```

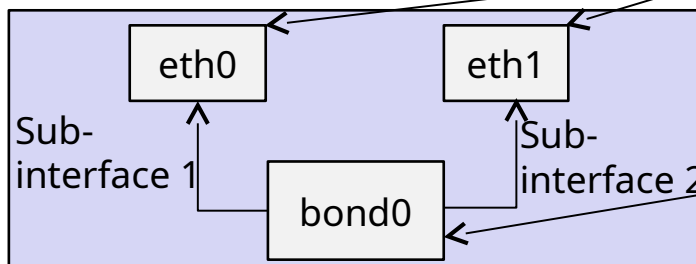
- ICMP echo request is transmitted in regular intervals via interface eth0 to a remote server with IP 9.9.9.9 (additional checks (e.g. test 10) are not shown in the example)
- If the check fails five times, unavailability of the interface is assumed

- Weighted random distribution load - balancing is applied per packet via the interfaces eth0 and eth1
- Scheduling ratio between eth0 and eth1 is 1:10 → e.g. eth0 is a 1 Gigabit interface and eth1 is a 10 Gigabit interface

Link aggregation / Bonding

- Capability to aggregate two or more interfaces to a logical one that can be used as a regular interface
- Goals (see also chapter 6.1.1 from IEEE Std. 802.1AX-2020)
 - **Increasing the bandwidth**
 - **Increasing availability**
 - **Load balancing**
- Terminology depends on context, e.g.
 - Linux: (Channel) Bonding
 - IEEE: Link aggregation
 - Cisco: Etherchannel

Bonding in Linux kernel



Physical interfaces

Virtual interface:

- MAC address of first/active sub-interface is assigned
- (Depending on mode:) All additional sub-interfaces use its MAC address

Link aggregation / Bonding support in the Linux kernel

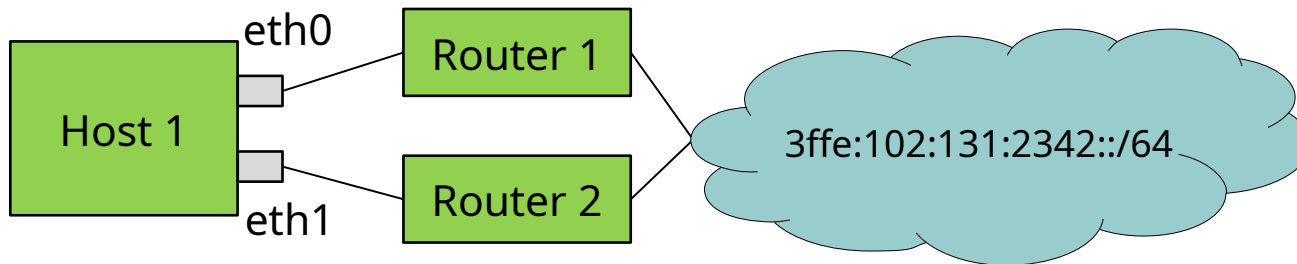
Configuration e.g. in */etc/network/interfaces*

```
auto bond0
iface bond0
    inet static address 10.0.1.42
    netmask 255.255.255.0
    network 10.0.1.0
    gateway 10.0.1.1
    slaves eth0 eth1
    bond_mode active-backup
    bond_miimon 100
    bond_downdelay 150
    bond_updelay 150
```

- Offers several modes, e.g.
 - *balance-rr*: Round-Robin starting with the first sub-interface
 - *active-backup*: only one sub-interface is active, in the case of an outage another interface is activated
 - *broadcast*: packets are transmitted via all sub-interfaces
 - *balance-tlb*: Interface is selected based on the load on the sub-interfaces
- Implementation in Linux source code available in the following folder (e.g. kernel version 6.17.3): */drivers/net/bonding*
- Implementation supports option to aggregate huge amounts of interfaces (no strict limit)

Equal-Cost Multi-Path Routing (ECMP)

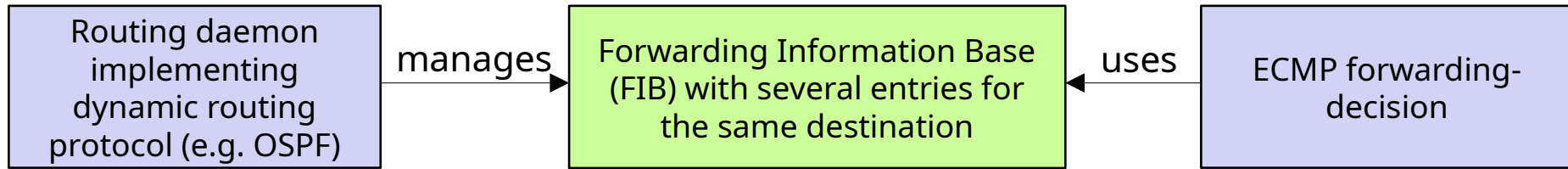
- FIBs may contain several entries for a specific destination (“next hops”)
- One of the potential next hops is selected when a forwarding decision is made
 - For an overview of the algorithm, see RFC 2992
- Next hop entry may be created as a static or dynamic (or direct) entry, e.g. adding two static entries using iproute2:



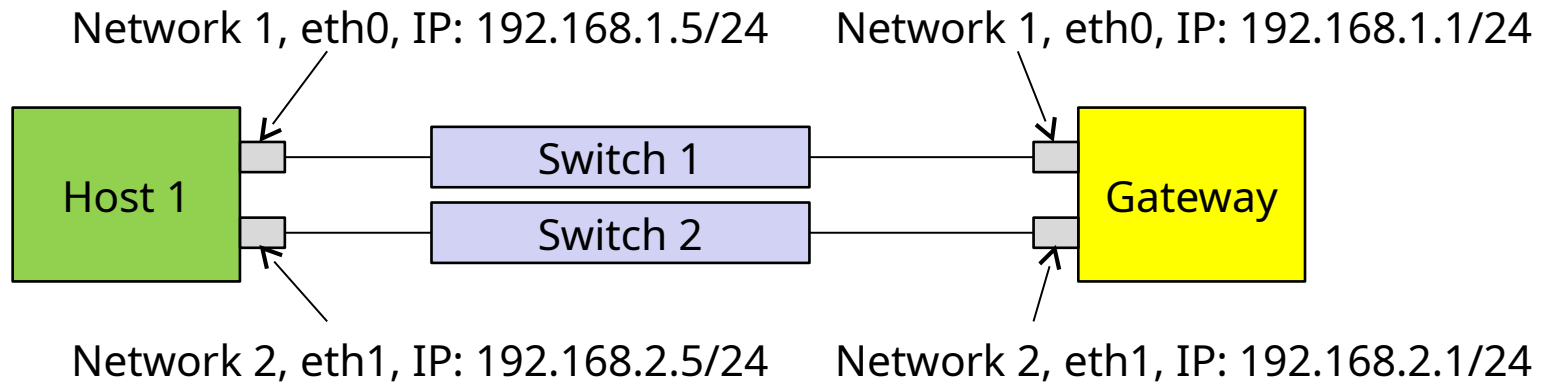
```
user@host$ ip -6 route add 3ffe:102:131:2342::/64 \  
    nexthop via fe80::211:1a2f:fea4:b2f1 dev eth0 \  
    nexthop via fe80::42:22fc:12ba:bba2 dev eth1
```

- Issue with the static definition of next hops:
 - In the case of a failure / down-time of any node on the path, the entry stays in the FIB → no fail-over

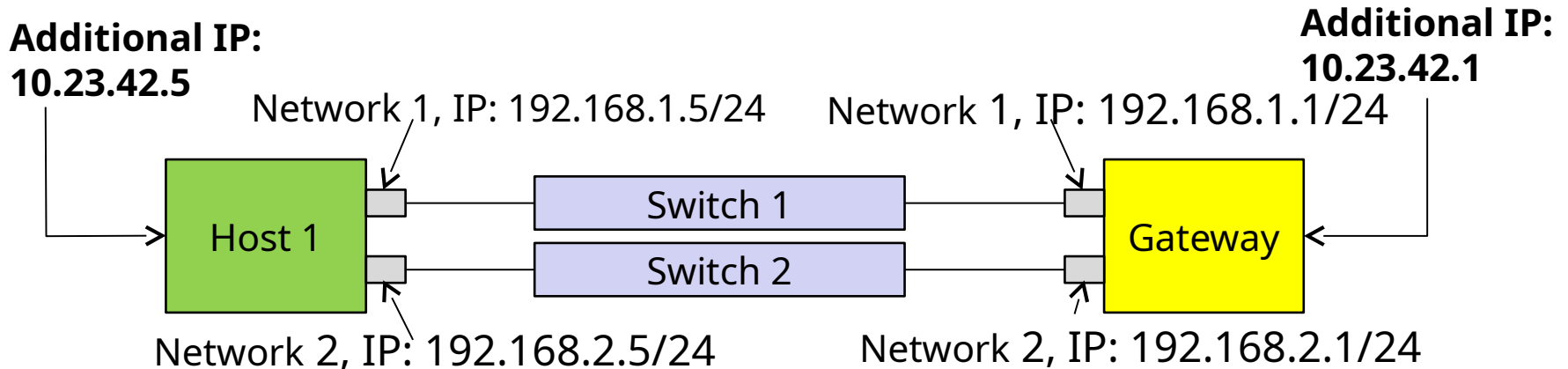
ECMP with OSPF



- Systems which have several alternative paths to a specific destination or a gateway, may manage paths via dynamic routing protocols for load balancing and fail-over purposes
- Required information is added to the FIB by routing daemons implementing these routing protocols
- Approach can e.g. be implemented using OSPF or IS-IS
- Protocol offering *unequal-cost* load balancing: Enhanced Interior Gateway Routing Protocol (EIGRP)
- Example scenario:



ECMP with OSPF



- **Approach:**

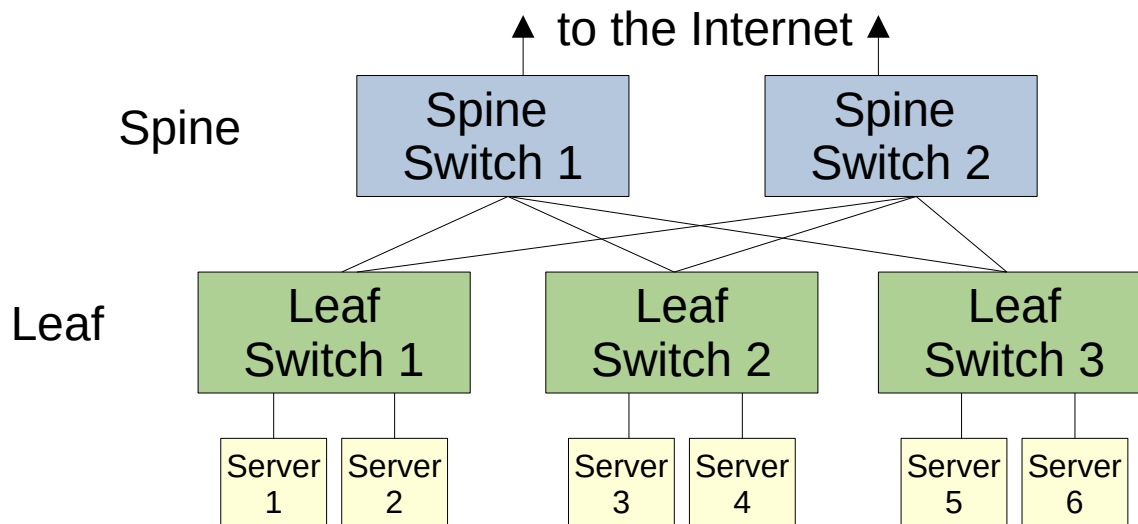
1. All routers/hosts accessible via alternative paths receive an additional IP which is independent from a physical interface
 - Assignment e.g. to a virtual interface / the loopback interface
2. IP is distributed using OSPF via all redundant paths
3. Participating systems learn available paths for reaching the additionally assigned IPs and add entries to their FIBs
 - Operating system cares for ECMP

- Example entry in the routing table of Host 1 in the scenario:

- Default Gateway 10.23.42.1 available via:
 - eth0: 192.168.1.1 **and** eth1: 192.168.2.1

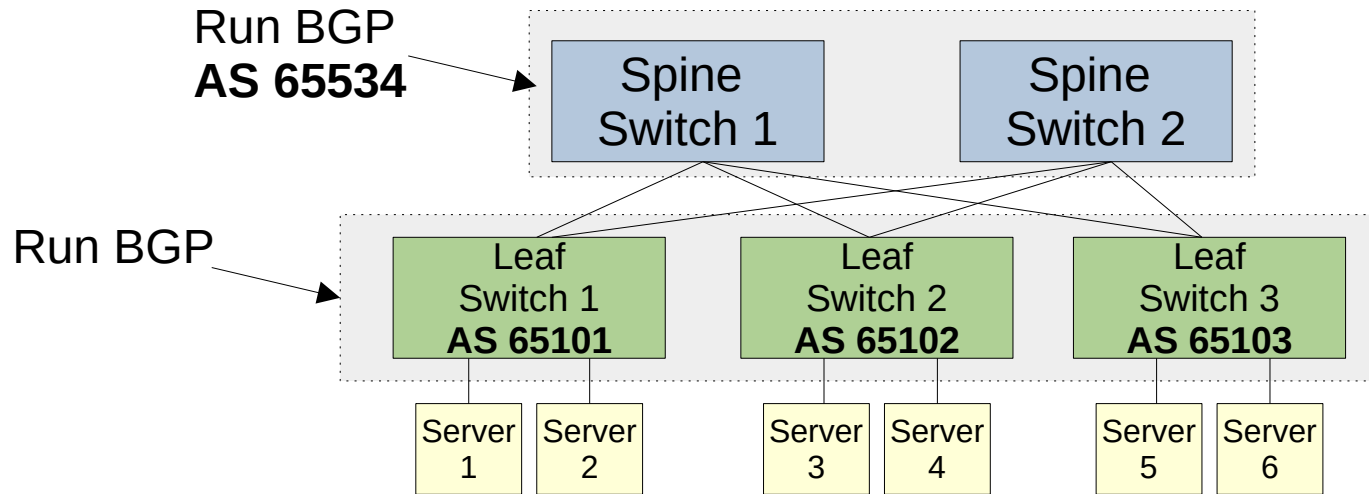
Excurses: Clos network topology

- L2 networks such as campus area networks were/are often structured into three segments:
 - core (providing the connection to e.g. the Internet),
 - aggregation (connecting core and access segments with each other) and
 - access segments (connecting the end systems such as servers)
- Due to limited scalability and fault-tolerance, this network architecture ("hierarchical internetworking architecture") is not used e.g. in modern datacenter infrastructures
- Instead, a **spine and leaf** architecture ("**Clos** architecture") is used



- 3-Stage folded Clos topology (Leaf→Spine→Leaf = 3 stages for server-to-server communication)
- With n -Port switches, $n^2 / 2$ servers are supported (assuming no oversubscription)
- $n + n/2$ switches are required

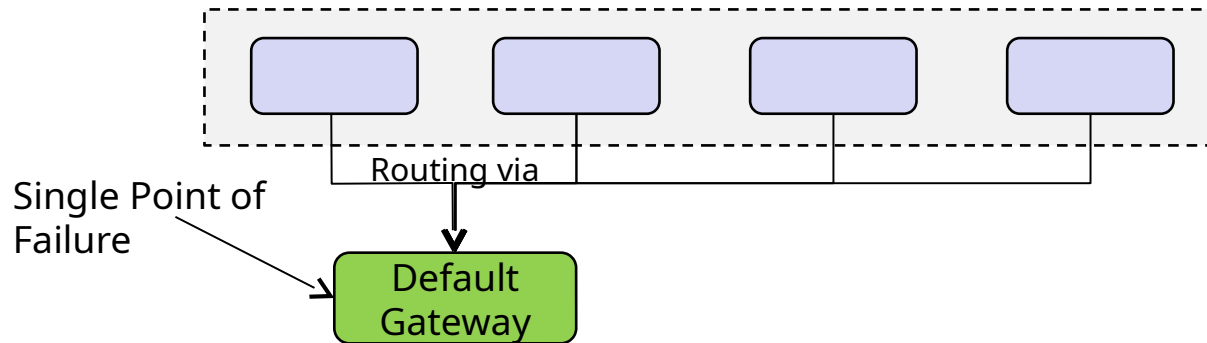
ECMP with BGP in a Clos network topology



- Server IPs in AS 65101: 192.168.1.0/24
- Server IPs in AS 65102: 192.168.2.0/24
- Server IPs in AS 65103: 192.168.3.0/24

- Spine layer forms a single Autonomous System (AS)
- Each leaf forms an own AS
- For server-to-server communication several paths exist → usage of ECMP
- Simple 3-stage folded Clos topology may be extended by several spine layers and clustering of leaf switches to scale the network
- BGP may also be used on servers (e.g. if virtualization / containerization is used on servers)
- See RFC 7938 for details of potential setups

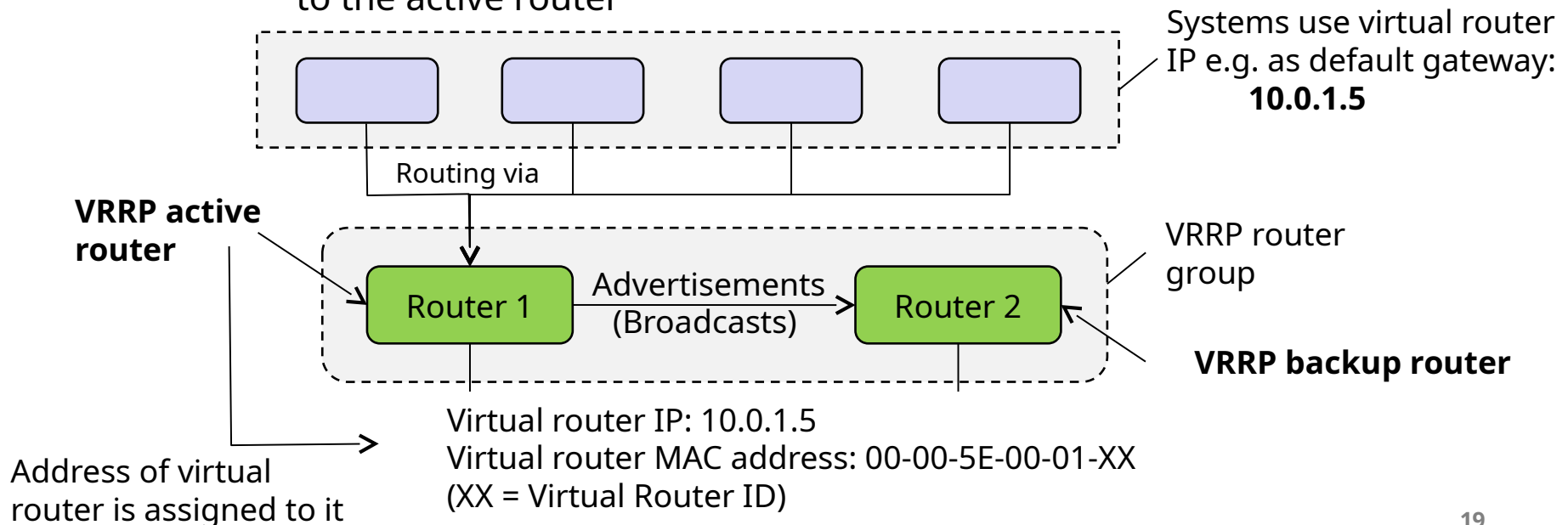
Redundancy of default gateways



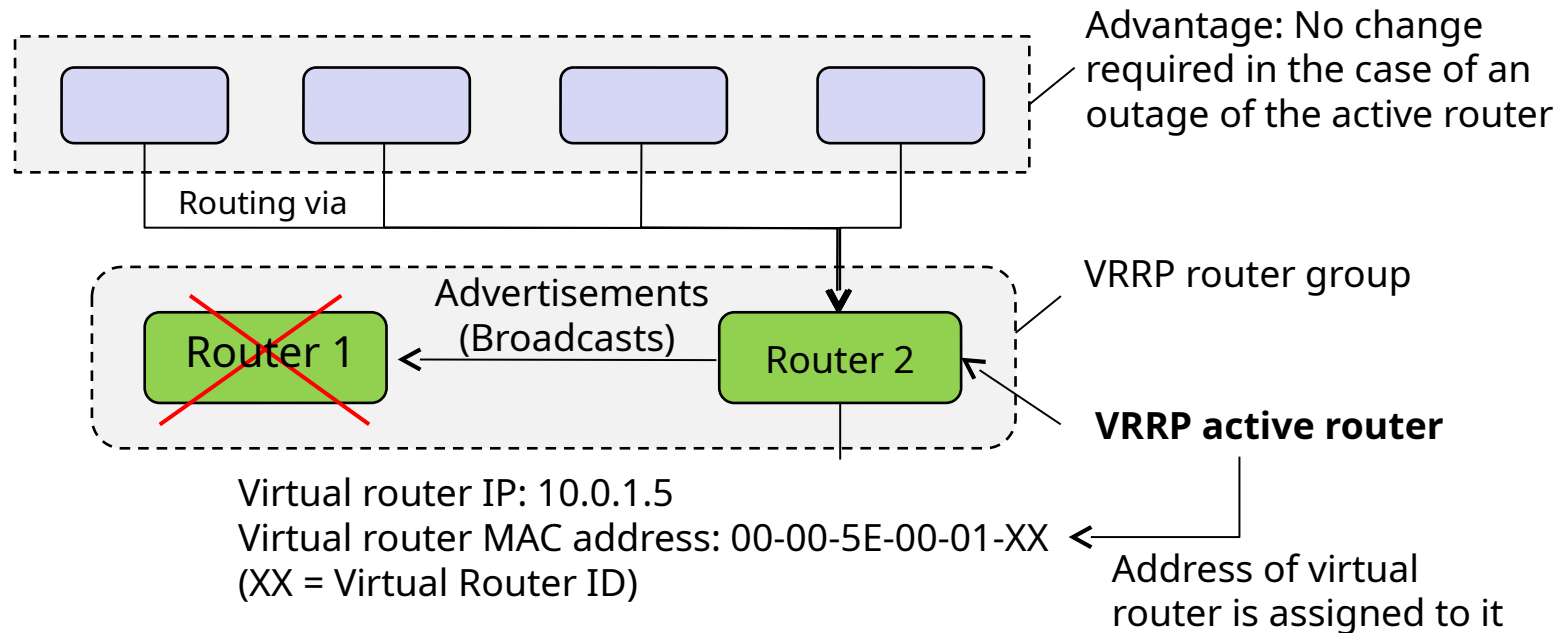
- Possible approaches:
 - Select default gateway from a set of routers via a dynamic routing protocol:
 - Drawback: Routing daemon is required on all systems, protocol overhead
 - IPv6 specific:
 - Discover router via the Neighbor Discovery Protocol
 - Drawback: If the router advertisement is not received anymore, it can not be derived from this fact that the router went down
 - Neighbor Unreachability Detection
 - Drawback: Detection of outages takes quite long
- Requirements:
 - Outage should be detected quickly
 - Protocol and management overhead should be small
 - Option for load balancing via redundant gateways should be provided

Virtual Router Redundancy Protocol (VRRP)

- Protocol defined in RFC 9568 (version 3) used to avoid single point of failure even if static route configuration is applied
- VRRP provides a mechanism to dynamically assign the role of a virtual router to one router in a set of routers
- Support IPv4 and IPv6
- Can be used for a broad spectrum of use cases (including high availability of services)
- Fundamental approach:
 - Router either has the role "active router" or "backup router"
 - Systems route packets via a "virtual router IP address" which is assigned to the active router

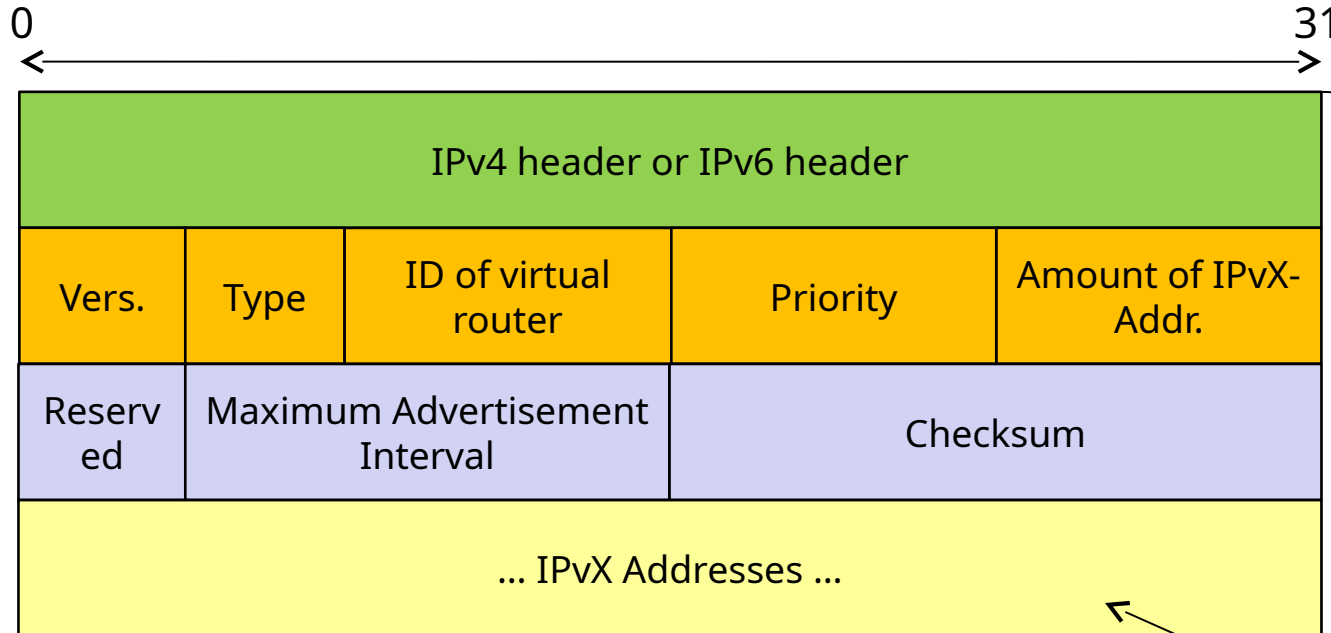


Virtual Router Redundancy Protocol (VRRP)



- If an advertisement is not received within a specific timer interval ("advertisement interval"), backup router may switch to the active router state
- New active router assigns virtual IP address and MAC address of the virtual router to its interface and cares for cache updates (via gratuitous ARP message / Neighbor Advertisement)
- Approach does not avoid issues if there are several backup routers that may switch to the active router state

VRRP – Protocol Details



- Packet is only accepted if it is transmitted within the same local network
→ TTL / Hop Limit = 255
- Packet is transmitted in regular intervals to a multicast address:
IPv4: 224.0.0.18
IPv6: ff02::12

IP addresses of the virtual routers

- Issue with a potential switch of several backup routers to the active role is solved based on priority assignment to routers:
 - $\text{Active_down_interval} = (3 * \text{advertisement_interval}) + \text{skew_time}$
 - $\text{Skew_time} = (((256 - \text{priority}) * \text{advertisement_interval}) / 256)$
→ router with higher priority detects down time of active router earlier
 - Additionally: If a router being the active router receives a packet with a higher priority than the own one, it will switch back to backup role

VRRP – Example configuration

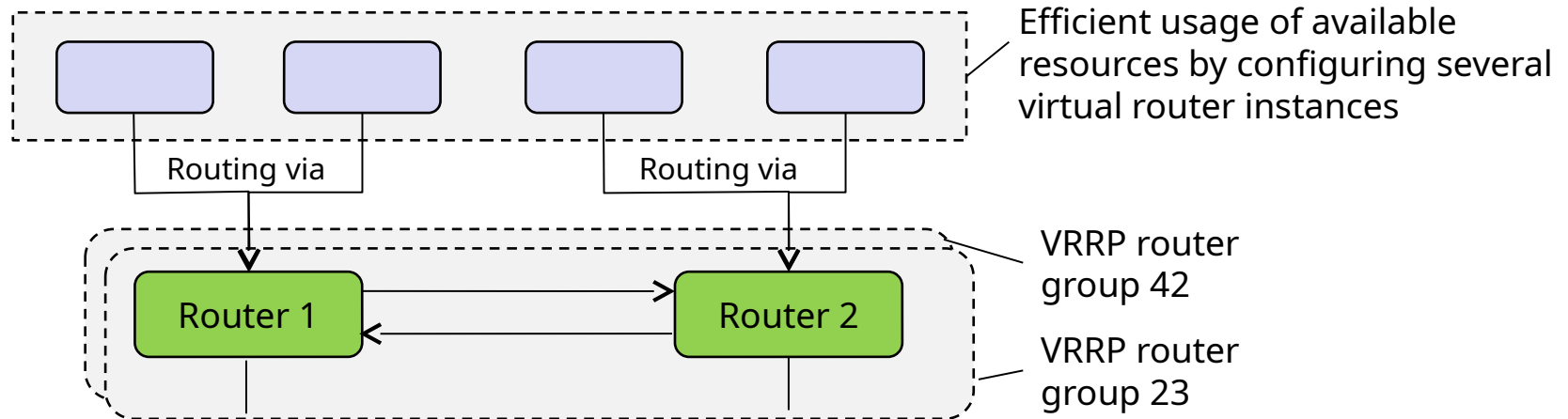
- Example configuration in a VyOs router (uses daemon *keepalived*) for router 1 for the below shown example:

```
user@Router1# set interfaces ethernet eth0 vrrp vrrp-group 42
user@Router1# set interfaces ethernet eth0 vrrp vrrp-group 42
virtual-address 10.0.1.5/8
user@Router1# set interfaces ethernet eth0 vrrp vrrp-group 42
preempt true
user@Router1# set interfaces ethernet eth0 vrrp vrrp-group 42
priority 200
```

- preempt (as specified in RFC 9568: Preempt_Mode):
 - "Controls whether a (starting or restarting) higher-priority Backup router preempts a lower-priority active router. Values are True to allow preemption and False to prohibit preemption. Default is True."
 - Preemptive mode can be ignored in the special case that the virtual IP is explicitly assigned to a specific router

VRRP – Load balancing scenario

- In a common VRRP setup, the backup routers are not used → available resources are not utilised



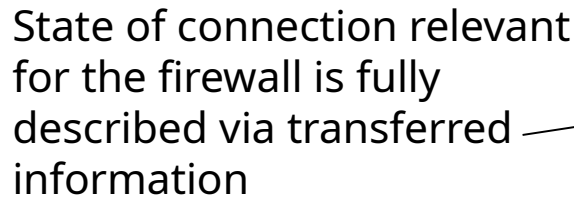
- Router 1: Active for virtual router 1, backup for virtual router 2

```
...  
user@Router1# set interfaces ethernet eth0 vrrp vrrp-group 42 priority 200  
user@Router1# set interfaces ethernet eth0 vrrp vrrp-group 23 priority 150
```

- Router 2: Active for virtual router 2, backup for virtual router 1

```
...  
user@Router2# set interfaces ethernet eth0 vrrp vrrp-group 42 priority 150  
user@Router2# set interfaces ethernet eth0 vrrp vrrp-group 23 priority 200
```

Firewall and
primary router
(e.g. active VRRP router)

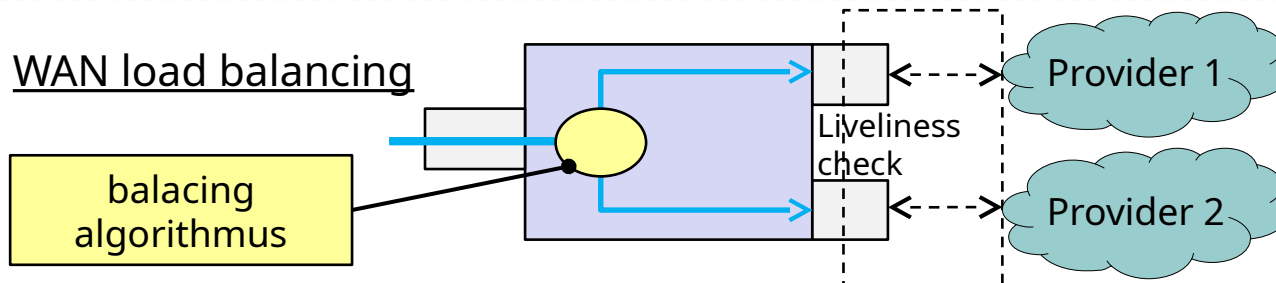
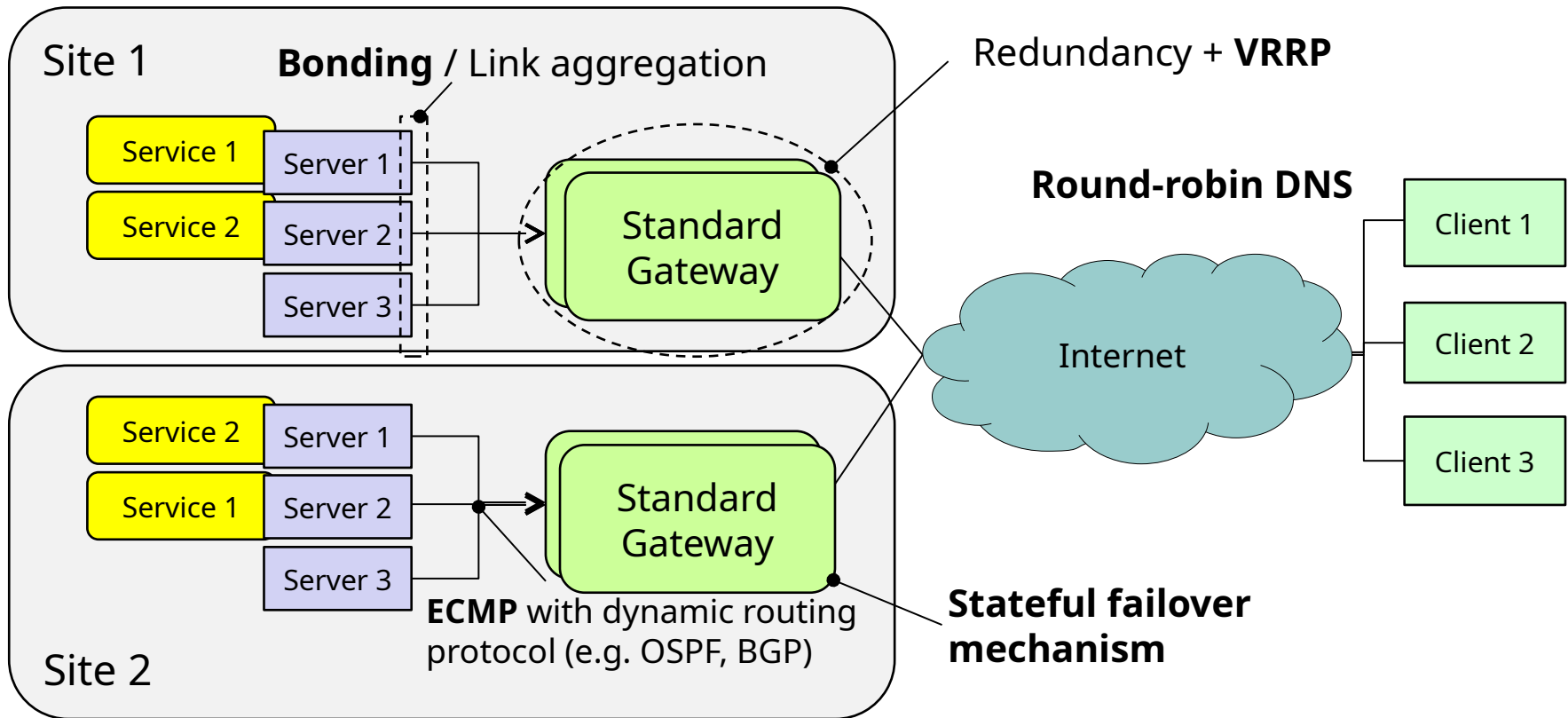


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Summary

- Goals: Maximize availability and use redundant resources continuously

$$\text{Availability} = \text{MTBF} / (\text{MTBF} + \text{MTTR})$$



References

Analysis of an Equal-Cost Multi-Path Algorithm <https://datatracker.ietf.org/doc/html/rfc2992>

BGP for Routing in Large-Scale Data Centers <https://datatracker.ietf.org/doc/html/rfc7938>

Virtual Router Redundancy Protocol (VRRP) <https://datatracker.ietf.org/doc/html/rfc9568>

Details about Bonding (Linux)
<https://www.kernel.org/doc/Documentation/networking/bonding.txt>